

Practice Quiz: Systems (Advanced Theory)

Name: __ Date: __

Part A: Multiple Choice

1. Variational free energy is defined as: A) The temperature of a system B) $F[q] = E_q[-\ln p(o, s)] + E_q[\ln q(s)] = \text{Energy} + \text{negative Entropy}$, an upper bound on negative log model evidence C) The total energy of a physical system D) The cost of computation
2. The Complexity minus Accuracy decomposition shows that: A) More complex models are always better B) Free energy minimization automatically balances model complexity (KL from prior) against data fit (expected log-likelihood) — implementing Occam's Razor C) Accuracy is irrelevant D) Complexity should be maximized
3. The mean-field approximation assumes: A) All variables interact B) The approximate posterior factorizes into independent groups: $q(s_1, s_2, \dots) = \prod_i q_i(s_i)$ C) Variables are deterministic D) The model is linear
4. On a factor graph, free energy decomposes into: A) A single global quantity B) Local free energies at each factor node, enabling distributed message-passing optimization C) Random components D) Physical energies
5. The Laplace approximation: A) Is always exact B) Approximates the posterior as a Gaussian centered at the MAP estimate — efficient but misses multimodality C) Uses sampling D) Requires infinite computation
6. When $q(s) = p(s|o)$ exactly: A) Free energy is maximized B) $F[q] = -\ln p(o)$ — free energy equals negative log evidence, and the KL approximation error is zero C) The model is wrong D) Entropy is zero
7. KL divergence's mode-seeking property ($D_{KL}[q||p]$) means: A) q spreads everywhere B) The variational approximation tends to "lock onto" one mode of p , potentially missing others C) q is always uniform D) KL is symmetric

Part B: Essay Questions

1. Derive the free energy functional from first principles for a general generative model $p(o, s)$. Show all three decompositions. For each decomposition, explain the computational significance — what does each term tell the brain (or the modeler) about the quality of inference? (500 words, include mathematical notation)
2. Compare mean-field variational inference, the Laplace approximation, and belief propagation (Bethe). For each: state the factorization assumption, derive the update equations, identify when it works well and when it fails. Which is most biologically plausible, and why? (500 words)
3. The free energy principle claims that all self-organizing systems minimize variational free energy. But variational free energy requires a generative model, an approximate posterior, and a definition of "observations." Critically examine: what is the justification for applying this mathematical framework to physical systems (not just brains)? Where does the framework break down at the boundaries? (400 words)